

Total No. of Questions : 4]

SEAT No. :

PD331

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**B.E. (Mechanical Engineering) (Insem)
ELECTRICAL AND HYBRID VEHICLE
(2019 Pattern) (Semester - VIII) (402051 E) (Elective-VI)**

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) Solve Q.1 or Q.2, Q.3 or Q.4.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Use of electronic pocket calculator allowed.
- 5) Assume suitable data, if necessary.

Q1) a) Explain the limitations of IC engine vehicles compared to electric vehicles. **[7]**

b) Explain exhaust emission and global warming due to IC engine vehicles and also discuss its effect on human health. **[8]**

OR

Q2) a) Describe overview of different EV challenges. **[7]**

b) Describe and illustrate any one case study of two wheeler electric vehicle available in Indian and global market. **[8]**

Q3) a) What is meant by 'gridable' HEV and explain its types. **[7]**

b) Explain with neat sketch construction and working of hybrid electric vehicle. Also state its advantages and disadvantages. **[8]**

OR

Q4) a) With a neat sketch, explain the configuration of Series hybrid electric drive train. Also state name of the vehicles in which this system is applied. **[8]**

b) What do you mean by tractive force? Explain the factors affecting the tractive force. **[7]**

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